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1 | INTRODUCTION

From confirmation reporrt We now consider settings in which the extremal dependence structure is non-stationary, and may change over time. For example, meteorological variables observed across spatial locations may exhibit changes in extremal dependence due to shifts in climate regimes, while biomarker measurements collected longitudinally for individuals may display structural changes associated with disease progression, treatment effects, or lifestyle changes (**????**). In such settings, it is of interest to detect changepoints or structural breaks in the extremal dependence structure.

In the existing framework, the multivariate "groups" could, in principle, represent different time periods. However, this method was not developed to accommodate temporally evolving dependence structures within an already multivariate setting (e.g., spatial), and we assumed that the data within each group was stationary. This is limiting in applications where tail dependence is expected to vary over time. In environmental contexts, for example, climate change may strengthen or weaken extremal dependence across an entire spatial domain, as suggested by **?**, who found that recent Southern European heatwaves have been more widespread than in the past. More generally, several types of structural change may occur. In some situations the extremal dependence structure may shift globally across all sites, corresponding to a regime change affecting the entire system. In other settings, only a subset of sites may change their extremal dependence behaviour, for example by switching between clusters of locations with similar tail dependence characteristics. Discarding the stationarity assumption and allowing for such temporal changes in extremal dependence is therefore a natural and important extension of our current work. It would enable us to identify time periods with distinct spatial patterns of tail dependence, as well as the specific changepoints at which these transitions occur. Therefore, a key focus of this work will be the development of methods for changepoint detection in multivariate extremal dependence structures, building on the divergence-based framework introduced in our first paper.

†This is an example for title footnote.
[0]**Abbreviations:** ANA, anti-nuclear antibodies; APC, antigen-presenting cells; IRF, interferon regulatory factor

There has been some previous work on changepoint detection in multivariate dependence structures, whether in the context of extremes or more generally. For example, **?** developed a non-parametric sliding window approach for detecting changepoints in pairwise extremal dependence. Also, **?** developed a Bayesian clustering algorithm for multivariate extremes which utilises a mixture of Dirichlet distributions, and they show how it can be used to temporally cluster negative returns for several S&P 500 stock sectors, thereby identifying periods of time with similar extremal dependence structures across the sectors. In **?**, a likelihood-based method for detecting temporal changepoints in spatio-temporal data is developed, and applied to daily wind speed data in Ireland, similar to the data we used in our first paper. We hope to build on these existing approaches and also the divergence-based clustering framework developed in our paper to develop a method for detecting changepoints in multivariate extremal dependence structures, which can be applied to a wide range of datasets and applications.

2 | METHODOLOGY

In this section, we describe a framework for detecting changepoints in multivariate extremal dependence structures. Section 2.1 introduces the CE framework of Heffernan and Tawn (2004) and **?**, which we use to model the extremal dependence structure of multivariate data. Section 2.2 describes how the extremal skew-geometric $\mathrm{JS}^{\mathrm{G}_\lambda}$ divergence can be used to quantify the difference between two extremal dependence structures. Finally, in Section 2.3, we describe a permutation-based procedure for detecting changepoints in the extremal dependence structure over time, using the divergence measure to quantify differences between temporal blocks.

2.1 | Conditional extremes model for multivariate extremal dependence

We model multivariate extremal dependence using the conditional extremes (CE) framework of Heffernan and Tawn (2004) and **?**. This framework describes the behaviour of a random vector conditional on one of its components exceeding a high threshold, and can accommodate both asymptotic dependence and asymptotic independence. We briefly summarise the spatial formulation used here; further details are provided in **?**.

Let $\boldsymbol{X}_s = (X_{1,s}, \dots, X_{d,s})$ denote a d-dimensional random vector observed at spatial location $s \in \{1, \dots, D\}$. Following **?**, we transform each component to standard Laplace margins, yielding $\boldsymbol{Y}_s = (Y_{1,s}, \dots, Y_{d,s})$. This standardisation separates marginal behaviour from the extremal dependence structure and permits both positive and negative associations.

For conditioning variable $i \in \{1, \dots, d\}$ and location $s \in \{1, \dots, D\}$, the CE model is

$$\left(\boldsymbol{Y}_{-i,s} \mid Y_{i,s} = y\right) = \boldsymbol{\alpha}_{|i,s} y + y^{\boldsymbol{\beta}_{|i,s}} \boldsymbol{Z}_{|i,s}, \qquad y > u_i, \tag{1}$$

where $\boldsymbol{Y}_{-i,s}$ denotes $\boldsymbol{Y}_s$ with its i^{th} component removed, u_i is a high threshold, and all vector operations are applied component-wise. The parameter vectors $\boldsymbol{\alpha}_{|i,s} \in [-1, 1]^{d-1}$ and $\boldsymbol{\beta}_{|i,s} \in (-\infty, 1]^{d-1}$ describe, respectively, the strength of extremal association and the conditional scaling of the remaining components. The residual vector $\boldsymbol{Z}_{|i,s}$ is assumed to be independent of $Y_{i,s}$ above u_i.

Following Heffernan and Tawn (2004), we adopt the working assumption

$$\boldsymbol{Z}_{|i,s} \sim \mathrm{MVN}_{d-1}\left(\boldsymbol{\mu}_{|i,s}, \Sigma_{|i,s}\right),$$

which gives

$$\left(\boldsymbol{Y}_{-i,s} \mid Y_{i,s} = y\right) \sim \mathrm{MVN}_{d-1}\left(\boldsymbol{\alpha}_{|i,s} y + y^{\boldsymbol{\beta}_{|i,s}} \boldsymbol{\mu}_{|i,s},\ \Omega_{|i,s}(y)\right), \qquad y > u_i, \tag{2}$$

where

$$\Omega_{|i,s}(y) = \mathrm{diag}\left(y^{\boldsymbol{\beta}_{|i,s}}\right) \Sigma_{|i,s}\, \mathrm{diag}\left(y^{\boldsymbol{\beta}_{|i,s}}\right).$$

The model is fitted separately for each conditioning variable, spatial location and temporal block, without imposing spatial smoothness. We set $u_i = u$, a high quantile of the standard Laplace distribution, for all i and s, to ensure that the resulting conditional distributions are comparable across locations, after Laplace standardisation. The Gaussian working model also permits the closed-form divergence calculations described in Section 2.2.

2.2 | Extremal skew-geometric Jensen-Shannon divergence for multivariate extremal dependence

To compare extremal dependence structures across spatial locations, we use the divergence-based framework of **?**. This approach measures dissimilarity between fitted CE models using the skew-geometric Jensen–Shannon divergence of **?**, which has a closed-form expression for the Gaussian conditional distributions in Equation (2).

For two distributions H and H^*, the dual skew-geometric Jensen–Shannon divergence is

$$\mathrm{JS}^{\mathrm{G}_\lambda}(H\|H^*) = (1-\lambda)\,\mathrm{KL}\left\{G_\lambda(H,H^*)\|H\right\} + \lambda\,\mathrm{KL}\left\{G_\lambda(H,H^*)\|H^*\right\}, \tag{3}$$

where KL denotes the Kullback–Leibler divergence and $G_\lambda(H,H^*)$ is the distribution with density proportional to the weighted geometric mean of the densities of H and H^*. We set $\lambda = 0.5$, yielding a symmetric, non-negative measure of dissimilarity. When H and H^* are multivariate Gaussian, Equation (3) can be evaluated in closed form; see **?** and **?** for details.

Let $H_{i,s,T}(y)$ denote the fitted conditional distribution of $\boldsymbol{Y}_{-i,s}$ given $Y_{i,s} = y$ at location s, estimated using observations from temporal block T. For two locations s and s^*, and conditioning variable i, the conditional extremal divergence is

$$\mathrm{cJS}^{\mathrm{G}_\lambda}_{i,s,s^*}(y;T) = \mathrm{JS}^{\mathrm{G}_\lambda}\left\{H_{i,s,T}(y)\|H_{i,s^*,T}(y)\right\}, \qquad y > u. \tag{4}$$

This quantity compares the fitted extremal dependence structures at the two locations for a specified value of the conditioning variable.

To remove the dependence on a particular conditioning value, we average Equation (4) over the distribution of threshold exceedances:

$$\mathrm{eJS}^{\mathrm{G}_\lambda}_{i,s,s^*}(T) = \mathbb{E}\left[\mathrm{cJS}^{\mathrm{G}_\lambda}_{i,s,s^*}(Y_i;T) \mid Y_i > u\right] = \int_u^\infty \mathrm{cJS}^{\mathrm{G}_\lambda}_{i,s,s^*}(y;T)\,h(y \mid Y_i > u)\,\mathrm{d}y, \tag{5}$$

where $h(y \mid Y_i > u)$ is the conditional density of a standard Laplace variable above u. As the marginal distribution and conditioning threshold are common across locations, this density does not depend on s, ensuring that the resulting expected divergence is comparable and symmetric across site pairs.

Needed? In practice, Equation (5) is approximated using a common Monte Carlo sample of conditioning values. To avoid unstable extrapolation beyond the observed range, the sample is truncated at a suitably high upper quantile of the standard Laplace distribution. Again, The same threshold u, taken to be a high quantile of the standard Laplace distribution, and Laplace samples are used across locations and temporal blocks, preventing additional Monte Carlo variation from being introduced into comparisons between blocks in our changepoint detection procedure in Section 2.3.

For each conditioning variable i, the expected divergences are collected in the dissimilarity matrix

$$M^{(i)}(T) = \left\{\mathrm{eJS}^{\mathrm{G}_\lambda}_{i,s,s^*}(T)\right\}_{s,s^*=1}^{D}, \tag{6}$$

where D is the number of spatial locations. Information across all conditioning variables can then be combined using the elementwise average

$$M(T) = \frac{1}{d}\sum_{i=1}^{d} M^{(i)}(T). \tag{7}$$

The matrix $M(T)$ summarises the spatial configuration of extremal dependence during temporal block T. Comparing matrices estimated from different blocks therefore allows us to assess whether this configuration changes over time, as described in Section 2.3.

2.3 | Changepoint detection in non-stationary multivariate extremal dependence structures

Let $M(T)$ denote the estimated extremal dependence structure for a time block T, which could represent a given period of time for which we have sufficient data to fit the CE model and estimate the corresponding divergence matrix. A simple way to compare two time blocks T_1 and T_2 is via the statistic

$$D_{\mathrm{obs}} := \|M(T_1) - M(T_2)\|_{\mathrm{F}}, \tag{8}$$

i.e. the Frobenius norm of their difference, defined as

$$\|A\|_{\mathrm{F}} := \sqrt{\sum_{i,j} A_{ij}^2}.$$

This statistic captures the overall or global magnitude of change in extremal dependence across all pairs of sites. Another option is the infinity, or maximum-entry, norm, defined as

$$\|A\|_\infty := \max_{i,j} |A_{ij}|,$$

which captures the largest difference between corresponding entries of the two matrices and therefore places greater emphasis on localised changes in extremal dependence. Critical values for this statistic could be generated via simulation from fitted models to assess whether observed changes represent significant shifts in tail dependence.

One key question is how large the observed statistic must be to indicate a significant change in extremal dependence. A natural way to assess whether the difference between two time periods reflects a genuine structural change, rather than sampling variability in the fitted CE models, is via a permutation test. A permutation test is a non-parametric method for testing hypotheses by comparing the observed test statistic to a reference distribution generated by randomly permuting the data (**?**). Such procedures are exact when the observations are exchangeable under the null hypothesis. One advantage of this approach is that it avoids strong parametric assumptions about the sampling distribution of the divergence matrix, which is unknown, and avoids specifying a parametric sampling distribution for the divergence matrix and can accommodate dependence-preserving permutation schemes.

2.3.1 | Permutation test for temporal changes in extremal dependence

Suppose we wish to test for a change in extremal dependence at a candidate split point τ by comparing two local time blocks immediately before and after the split. Let T_1 and T_2 denote the corresponding pre- and post-change windows, each of size n. We fit the CE model separately in each block, yielding divergence matrices $M(T_1)$ and $M(T_2)$ as defined in Equation (6) (or their aggregate form in Equation (7)).

Our null hypothesis is

$$H_0 : \text{ the extremal dependence structure is identical in } T_1 \text{ and } T_2.$$

Under this null hypothesis of stationarity, exceedances from the two blocks arise from the same distribution. We therefore generate a reference distribution for D_{obs} by repeatedly permuting the time indices:

1. **Pool observations:** For a chosen block size $n \le \min\{|T_1|, |T_2|\}$, combine the final n observations from time block T_1 and the first n observations from T_2 into a single dataset.

2. **Permute temporal labels:** Randomly reassign the pooled observations, or appropriate dependence-preserving temporal groups, to two pseudo-blocks of size n.

3. **Refit CE models:** For the permuted dataset, reapply the exceedance threshold u for each conditioning variable and refit the CE model in each pseudo-block, yielding divergence matrices $M^{(b)}(T_1)$ and $M^{(b)}(T_2)$, for $b = 1, \dots, B$ permutations.

4. **Recompute statistic:** Compute the test statistic

$$D^{(b)} := \left\| M^{(b)}(T_1) - M^{(b)}(T_2) \right\|_{\mathrm{r}}, \qquad b = 1, \dots, B, \tag{9}$$

for the r-norm of choice (Frobenius or Infinity). As the CE model fitting and divergence calculation are repeated within each permutation, this procedure naturally propagates the estimation uncertainty in the fitted dependence parameters, somewhat analogous to a bootstrapping approach. Remove reference to bootstrap?

The empirical p-value is then

$$\hat{p} = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\left\{ D^{(b)} \ge D_{\text{obs}} \right\}}{B+1}. \tag{10}$$

A small value of $\hat{p}$ provides evidence that the tail-dependence structure differs between the two time periods. In performing the permutation test, a suitable block size n must be chosen to ensure reliable estimation of the CE model parameters within each time block, while also providing sufficient power to detect changes in extremal dependence. Using the same n across all permutations ensures that the sampling variability in the fitted parameters is appropriately reflected in the reference distribution of the test statistic.

2.3.2 | Sliding-window permutation approach for changepoint localisation

The permutation test above assesses whether a *given* time split corresponds to a change in extremal dependence. We now extend this procedure across a sequence of candidate split points to identify periods at which the extremal dependence structure changes.

Given N ordered temporal units and a block size n, consider candidate changepoints $\tau \in \{n, \dots, N-n\}$. For each candidate τ, define the adjacent temporal blocks

$$T_1(\tau) := \{\tau - n + 1, \dots, \tau\}, \qquad T_2(\tau) = \{\tau + 1, \dots, \tau + n\}. \tag{11}$$

The candidate τ therefore represents a possible change between temporal units τ and $\tau + 1$.

We first evaluate the observed discrepancy

$$D(\tau) = \left\| M\{T_1(\tau)\} - M\{T_2(\tau)\} \right\|_{\mathrm{r}} \tag{12}$$

over the admissible candidate split points, for $r \in \{\mathrm{F}, \infty\}$. The resulting profile provides an initial screening of temporal changes in the estimated extremal dependence structure. In particular, local maxima of $D(\tau)$ identify candidate regions where the fitted dependence structures on either side of the split differ most strongly.

We then apply the permutation test described above at each feasible candidate τ, yielding a corresponding profile of pointwise p-values,

$$\{\hat{p}(\tau) : \tau \in \{n, \ldots, N - n\}\} .$$

A small value of $\hat{p}(\tau)$ indicates that the observed difference between the extremal dependence structures in $T_1(\tau)$ and $T_2(\tau)$ is larger than expected under temporal homogeneity. Examining the screening statistic and permutation p-values together distinguishes pronounced estimated changes from changes that may instead be attributable to sampling variability.

precipitation or drought data?? When the number of admissible split points is modest, as in our application to Spanish temperature and precipitation data in Section 4, permutation testing can be performed over the complete candidate range. In settings with longer time series or substantially larger spatial networks, the screening profile may instead be used to restrict permutation testing to neighbourhoods around its most prominent local maxima, thereby reducing computational cost.

As changes in environmental dependence structures may occur gradually, a candidate τ should not necessarily be interpreted as the time of an instantaneous physical transition. Rather, it identifies a division between adjacent periods whose estimated extremal dependence structures differ.

3 | SIMULATION STUDY

We demonstrate the ability of the proposed procedure to detect temporal changes in multivariate extremal dependence through a simulation study. The simulation design broadly reflects the structure of the Spanish application in Section 4, with two variables observed weekly at multiple sites over a sequence of "season-years" (defined as the period from October to September). Rather than imposing an instantaneous changepoint, we allow the extremal dependence structure to evolve gradually over a specified transition period, to better reflect the types of changes that may occur in environmental systems. The objective is therefore not to recover a single exact change time, but to assess whether the procedure identifies evidence of a transition and whether the Frobenius and infinity norms respond differently to widespread and localised changes.

In Section 3.1, we describe the simulation setup and the scenarios considered, which include a null scenario with no change, a local change affecting a subset of sites, and a global change affecting all sites. TODO: Describe results section here

3.1 | Simulation setup

We simulate two variables at $D = 40$ sites over 60 season-years, with 12 weekly observations in each season-year. At each site and time point, the two variables are generated from a bivariate Student t-copula with three degrees of freedom (**?**). The sites are generated independently, and the weekly observations are conditionally independent given the site- and year-specific copula parameters. The observations are transformed to standard Laplace margins using their known marginal distributions. Across all CE fits, the same threshold u, set to the 85th quantile of the standard Laplace distribution, as well as 500 Monte Carlo samples of the conditioning variable, are used to evaluate the expected divergence in Equation (5), to ensure comparability of the resulting conditional distributions.

Need for mathematical notation/description? The sites are divided into three baseline extremal dependence groups. Sites $1, \ldots, 14$ have initial t-copula correlation parameter $\rho_{s,0} = 0.1$, sites $15, \ldots, 27$ have $\rho_{s,0} = 0.5$, and sites $28, \ldots, 40$ have $\rho_{s,0} = 0.8$. These groups produce heterogeneous extremal dependence across the spatial domain before any temporal change is introduced.

For an affected site s, we allow the t-copula correlation parameter to evolve according to

$$\rho_{s,t} := \tanh \operatorname{atanh}(\rho_{s,0}) + A_s \delta_z g(t) , \tag{13}$$

where A_s indicates whether site s is affected by the change, δ_z controls its magnitude on the Fisher-z scale, and $g(t)$ determines its temporal progression. Specifically,

$$g(t) = \begin{cases} 0, & t \leq t_0, \\ \dfrac{t - t_0}{t_1 - t_0}, & t_0 < t < t_1, \\ 1, & t \geq t_1. \end{cases} \tag{14}$$

so that the dependence parameter is constant before season-year t_0, changes gradually between t_0 and t_1, and remains constant thereafter. The transformations $\operatorname{atanh}(\cdot)$ and $\tanh(\cdot)$ allow the change to be specified using an unrestricted linear predictor while ensuring that $\rho_{s,t} \in (-1, 1)$ throughout.

We set $\delta_z = 0.45$, under which an affected site with baseline correlation $\rho_{s,0} = 0.1$ reaches a final correlation of approximately 0.5. For the other two baseline groups, the corresponding final correlations are approximately 0.76 and 0.91. We consider three scenarios:

(i) **Null:** no sites are affected, so that $A_s = 0$ for all s;

(ii) **Local change:** five sites from the first baseline group are affected, so that $A_s = 1$ for $s = 1, \dots, 5$ and $A_s = 0$ otherwise;

(iii) **Global change:** all 40 sites are affected, so that $A_s = 1$ for every s.

The local scenario represents a concentrated change in the extremal dependence structure, whereas the global scenario represents a broader system-wide transition. We therefore expect the infinity norm to be particularly responsive to the local alternative and the Frobenius norm to benefit from the accumulation of pairwise changes under the global alternative.

For each scenario, we generate 100 independent datasets and apply the changepoint procedure described in Section 2.3. For the screening step, we fit using adjacent blocks of $15, 20, 25$ and 30 season-years, and evaluate candidate split points over the complete feasible range. The CE model is fitted using adjacent blocks of 25 season-years, which was identified as optimal, in terms of balancing estimation accuracy and power to detect changes, for our application in Section 4, and candidate split points are evaluated over the complete feasible range. For each replicate and norm, we record the screening statistic $D(\tau)$ and the corresponding permutation p-values $\hat{p}(\tau)$ for each candidate split point τ. Under the null scenario, the p-values give an empirical frequency with which the procedure identifies an apparent change in a temporally stationary dependence structure. Under the local and global alternatives, they measure the ability of the procedure to detect the simulated transition and identify the region to which the estimated change is localised.

3.2 | Simulation results

- We simulate weekly observations for one season at 40 locations over 60 years, with 12 observations per season–year, i.e. for a single “season”, as found in our application section in Section 4.
- At each location s and year t, the two variables are generated from a bivariate t-copula with $\nu = 3$ degrees of freedom and known generalized Pareto marginal distributions.
- The locations are divided into three baseline dependence clusters: 14 locations with $\rho_{s,0} = 0.2$, 13 with $\rho_{s,0} = 0.5$, and 13 with $\rho_{s,0} = 0.8$.
- A gradual change in the t-copula correlation parameter is introduced using

$$\rho_{s,t} = \tanh\left\{\operatorname{atanh}(\rho_{s,0}) + A_s \delta_z g(t)\right\},$$

  where A_s indicates whether location s is affected, δ_z controls the magnitude and direction of the change, and $g(t)$ increases linearly from 0 to 1 over the specified transition period.
- The transformation $\operatorname{atanh}(\rho_{s,0})$ maps the baseline correlation from $(-1, 1)$ to the unrestricted real line. The change is applied on this Fisher-z scale, after which $\tanh(\cdot)$ maps the linear predictor back to $(-1, 1)$, ensuring that the resulting copula correlation remains valid.
- We consider a local-change setting, in which $A_s = 1$ for only a selected subset of locations, and a global-change setting, in which $A_s = 1$ for all locations.

4 | APPLICATION TO SPANISH DATA

- We now apply our changepoint detection algorithm to data from Spain.
- In Section 4.1, we introduce the dataset of temperatures and precipitation indexes used in our application.
- Section 4.3 shows the results of our initial screening step, while in Section 4.4, we discuss the results of our changepoint detection algorithm.
- Finally, Section 4.5 shows the clustering results for seasons before and after identified changepoints.

4.1 | Spanish Data

- We analyse temperature and precipitation observations from 40 meteorological stations distributed across mainland Spain.
- Their locations are shown in the left panel of Figure 1 .
- The analysis covers the period 1960-2020, inclusive.
- Only stations satisfying the required record-length and data-availability criteria are retained. State criteria for choosing stations/locations (description in code).
- Daily maximum temperature and precipitation totals are obtained from the European Climate Assessment Dataset (ECAD) (European Climate Assessment & Dataset (ECA&D) n.d.).
- The ECAD provides quality-controlled daily meteorological observations from stations across Europe.
- We aggregate the daily observations to a weekly resolution, taking the maximum recorded temperature and total accumulated precipitation within each week.
- This aggregation reduces the influence of very short-range temporal variation and decreases the proportion of zero precipitation observations, although it does not necessarily remove temporal dependence completely.
- The relationship between high temperature and dry conditions may depend more strongly on precipitation accumulated over preceding weeks than on precipitation observed during the same week.
- We therefore represent precipitation conditions using the Standardised Precipitation Index (SPI) of McKee, Doesken, and Kleist (1993), rather than using weekly precipitation totals directly.
- At each station, we calculate SPI from precipitation accumulated over a rolling 13-week window.
- Thus, the value assigned to a given week summarises precipitation during that week and the preceding 12 weeks, providing a measure of precipitation conditions over an approximately three-month period.
- The accumulated precipitation is modelled using a Gamma distribution, whose parameters are estimated using unbiased probability-weighted moments.
- Its fitted cumulative probability is then transformed to a standard Gaussian scale.
- Under the usual SPI convention, negative values represent drier-than-normal conditions and positive values represent wetter-than-normal conditions.
- For the dependence analysis, we therefore define the drought index [D=-SPI,] so that larger values correspond to drier conditions.
- This sign convention means that high temperatures and dry conditions are expected to exhibit positive extremal dependence.
- The right-hand panels of Figure 1 illustrate the relationship between weekly maximum temperature and the drought index at Daroca, in Aragon, and Jerez de la Frontera, in Andalusia.

- These stations are highlighted in the map in the left-hand panel.
- At both stations, the distribution of weekly maximum temperature differs substantially between seasons.
- The seasonal separation in drought-index values is less pronounced at Daroca, whereas Jerez de la Frontera exhibits clearer seasonal variation in both temperature and drought conditions. Mention explicit figures for means/95th quantiles??
- These seasonal differences motivate fitting the dependence model and applying the changepoint procedure separately within each season.
- This allows both the marginal behaviour and the extremal dependence structure to vary seasonally.

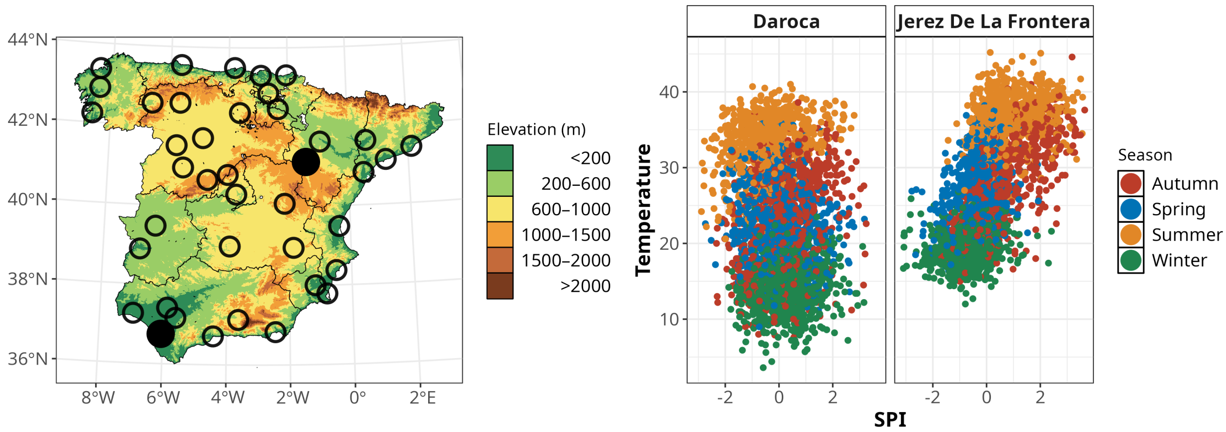

FIGURE 1 Left: Elevation (m) profile of mainland Spain along with locations of 40 meteorological stations. Daroca and Jerez de la Frontera highlighted. Right: Weekly maximum temperature against 13-week drought index, (D=-SPI), at these two stations, with observations coloured by season.

4.2 | Setup

- The conditional extremes framework is applied after transforming each variable to a common marginal distribution.
- Consequently, systematic changes in the marginal distributions over time could otherwise be confounded with changes in the extremal dependence structure.
- Both variables exhibit seasonal variation, while weekly maximum temperature in particular also displays a long-term marginal trend.
- We therefore preprocess the marginal distributions before fitting the conditional extremes models.
- At each station, and for each variable, we consider observations from each calendar month separately and fit a linear trend in year.
- The estimated trend is removed from the observations for that month.
- Treating months separately allows the magnitude of the temporal trend to vary throughout the year while accounting for the pronounced marginal seasonality.
- This detrending removes an estimated linear change in the location of each monthly marginal distribution; it does not imply that all temporal dependence has been removed.

- In particular, some short-range dependence may remain between successive weekly observations, and consecutive values of the 13-week drought index are constructed from overlapping precipitation windows.
- Following detrending, the observations are grouped by meteorological season.
- For each station and variable, the resulting seasonal observations are transformed standard Laplace margins using the empirical rank transform.
- The conditional extremes and changepoint analyses are then performed separately for winter, spring, summer and autumn.
- We fit the conditional extremes model of Heffernan and Tawn (2004) using a common dependence threshold equal to the theoretical 80th percentile of the standard Laplace distribution.
- The same threshold is used for both variables, at every station, in both temporal blocks and at every candidate changepoint.
- This ensures that exceedances have a common interpretation throughout the analysis.
- The 80th-percentile threshold provided a satisfactory empirical compromise between restricting attention to the joint tail and retaining enough exceedances for stable model estimation.
- In particular, the resulting clustering solutions exhibited clear spatial contiguity. Add a sensitivity analysis for the dependence threshold.
- The Monte Carlo approximation used to construct the pairwise dissimilarities employs the same sample from the standard Laplace distribution in every temporal comparison.
- Holding this sample fixed avoids introducing additional Monte Carlo variation between candidate changepoints.
- The conditional fits from the two conditioning directions are combined into the aggregated dissimilarity matrix defined in Section **??**. Test for others
- All screening, testing and clustering calculations reported below use this matrix. Confirm terminology and add sensitivity results for the alternative dissimilarity matrices, if required.
- For the Spanish application, the permutation procedure is adapted to preserve the temporal and spatial structure within each season.
- Rather than permuting individual weekly observations, we permute complete season-years.
- For spring, summer and autumn, the season-year is the calendar year in which the observations occur.
- Winter spans two calendar years; December is therefore assigned to the same season-year as the following January and February.
- At a candidate season-year (t), the two temporal blocks consist of the (25) season-years ending at (t) and the (25) season-years beginning at ($t + 1$), respectively.
- Thus, (t) denotes a possible change between season-years (t) and ($t + 1$).
- In each pointwise permutation, the 50 season-years from the two blocks are pooled and randomly divided into two pseudo-blocks containing 25 season-years each.
- Every weekly observation belonging to a given season-year is moved as a single unit, and the same reassignment is applied simultaneously to both variables and all 40 stations.
- This preserves the dependence between observations within a season-year and the spatial correspondence between stations.
- The validity of this permutation procedure relies on treating season-years as exchangeable under the null hypothesis of no change in the extremal dependence structure.

- Although permuting complete season-years preserves within-year dependence, it does not preserve any residual dependence between successive season-years.
- The choice of block length involves a bias-variance trade-off.
- Shorter blocks provide better temporal localisation but contain fewer threshold exceedances, resulting in unstable or failed conditional extremes fits.
- Longer blocks provide more stable estimates but leave fewer feasible candidate changepoints and smooth changes over a wider temporal interval.
- We investigated block lengths of 15, 20, 25 and 30 season-years.
- With 15-year blocks, many candidate comparisons failed because at least one conditional fit contained too few exceedances.
- Twenty-year blocks were reliable for winter but produced substantial failure rates for several other seasons.
- All candidate comparisons fitted successfully with 25-year blocks.
- Increasing the length to 30 years also produced stable fits, but left only two to five candidate changepoints per season.
- We therefore use 25 season-years on either side of each candidate throughout the primary analysis.
- This was the shortest common block length that produced stable fits across all seasons while retaining a useful range of candidate changepoints. Present the alternative block lengths as a sensitivity analysis or in the supplementary material(?).

4.3 | Screening

- We first perform a computationally less intensive screening step, in which the observed discrepancy between the two estimated dissimilarity matrices is calculated at every feasible candidate season-year without generating a permutation distribution.
- Figure 2 presents the screening statistics for each season using the Frobenius, infinity and spectral norms.
- As discussed in Section 2, these norms provide complementary summaries of the difference between the dissimilarity matrices on either side of a candidate change.
- The three norms have different numerical scales, and their absolute values should therefore not be compared directly.
- Instead, we compare the timing and relative prominence of features in their respective screening curves.
- Red points in Figure 2 identify local peaks.
- For a given norm (T), an interior candidate season-year (t) is classified as a local peak when

  $$T(t) > T(t-1) \qquad \text{and} \qquad T(t) \geq T(t+1).$$

  The first and final candidate years are not classified as local peaks because they cannot be compared with candidates on both sides.
- For winter, all three norms increase during the late 1990s and exhibit a prominent shared peak in 1998.
- A smaller common peak is present around 1992.
- The agreement among the norms makes the late-1990s period a particularly notable candidate region.
- For spring, the clearest screening peak occurs around 1992, for all norms.
- Smaller local peaks occur around 1989 and 1997.

- For summer, the screening curves contain several prominent peaks.
- The largest norm values occur in 1988, with further substantial peaks in 1992 and 1997.
- For Autumn, the screening results are less concentrated around a single candidate year.
- The Frobenius curve has local peaks around 1990 and 1996, while the infinity and spectral curves identify broadly similar candidate regions during the late 1980s or early 1990s and the mid-1990s.
- The precise locations of the peaks are less consistent across norms than for winter, spring or summer.
- Across the four seasons, the largest Frobenius norm screening value occurs for summer around 1988. Cite exact value
- The clearest agreement among all three norms occurs around 1992 for spring and around 1998 for winter.
- Although the screening step highlights several candidate regions of particular interest, the total number of feasible candidate season-years is sufficiently small to apply the permutation test at every candidate.
- We therefore retain the complete candidate range rather than restricting the formal analysis to the local screening peaks.

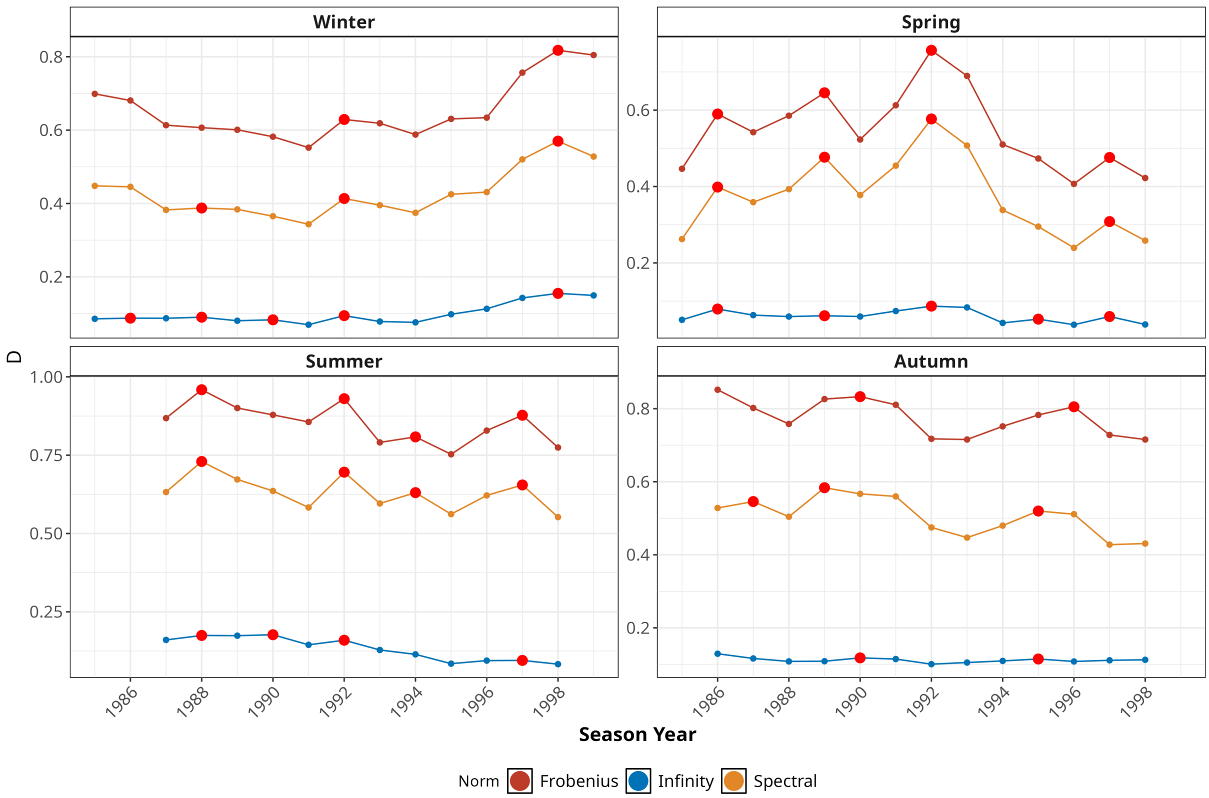

FIGURE 2 Screening discrepancies between the estimated extremal-dependence dissimilarity matrices on either side of each candidate season-year, calculated using 25 season-years per temporal block. Results are shown for the test statistic D using the Frobenius, infinity and spectral norms. A candidate (t) represents a possible change between season-years (t) and $(t+1)$. Red points denote interior local peaks satisfying $(T(t) > T(t-1))$ and $(T(t) \geq T(t+1))$.

4.4 | Changepoint detection

- We next apply the permutation test at every feasible candidate season-year.
- For each candidate and norm, the null distribution is estimated using 1000 permutations of complete season-years between the two 25-year blocks.
- Figure 3 shows the resulting pointwise Monte Carlo p-values for the Frobenius, infinity and spectral norms.
- A candidate season-year (t) represents a possible change between season-years (t) and $(t + 1)$, and the horizontal dashed line denotes the pointwise 5% significance level.
- The p-values in Figure 3 are pointwise: each tests a change at one specified candidate season-year.
- For spring, none of the three norms produces a pointwise p-value below 0.05.
- The smallest values occur around 1992, in agreement with the dominant feature identified during screening, but remain above the pointwise significance threshold.
- Also for Autumn, none of the three norms produces a pointwise p-value below 0.05, although the infinity norm came very close for 1986, the year with the highest test statistic in our screening step.
- For winter, the infinity norm provides pointwise evidence of a change over the period 1996–1999.
- The spectral norm supports a narrower region around 1997–1998, while the Frobenius norm is significant in 1998.
- Thus, all three norms identify 1998, and the overall pattern provides consistent evidence for a change in the winter extremal dependence structure during the late 1990s.
- For summer, the infinity norm produces pointwise-significant p-values over an extended period from approximately 1987 to 1993.
- The Frobenius and spectral norms support narrower subsets of this region, with particularly small p-values around 1988-1990 and 1992.
- The agreement among the norms indicates that the early-1990s signal is not confined to a single choice of discrepancy statistic.
- After approximately 1994, the summer p-values generally increase, particularly for the infinity norm.
- These results provide no pointwise evidence for a change at these later candidates.
- However, the relatively large p-values should not be interpreted as affirmative evidence that the dependence structure is unchanged; they indicate only that the observed discrepancies are not unusual under the corresponding permutation null distributions.
- The infinity norm identifies broader pointwise-significant candidate ranges than the Frobenius and spectral norms for both summer and winter.
- As discussed in Section **??**, this may indicate that some changes are more pronounced in particular entries of the dissimilarity matrix than across its entire structure.
- Nevertheless, the principal summer and winter candidate regions are supported by more than one norm.
- In particular, all three norms provide pointwise evidence around 1990 for summer and around 1998 for winter.
- Taken together, the results suggest candidate changes in the summer extremal dependence structure during the late 1980s or early 1990s and in the winter structure during the late 1990s.
- No comparable evidence is found for spring or autumn.

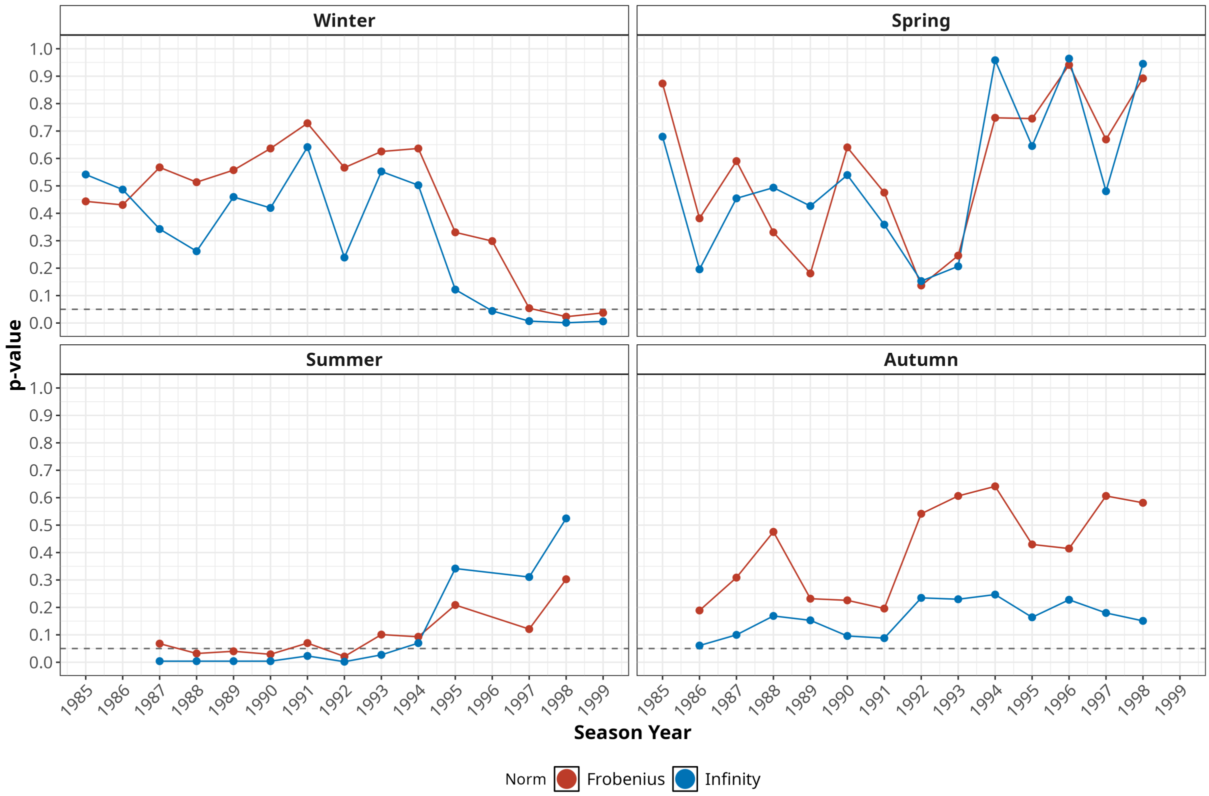

FIGURE 3 Pointwise permutation-test p-values for candidate changes in the seasonal extremal dependence structure, calculated using 25 season-years per temporal block and 200 permutations at each candidate. Results are shown for the Frobenius, infinity and spectral norms. A candidate (t) represents a possible change between season-years (t) and ($t + 1$). The horizontal dashed line denotes the pointwise 5% significance level.

4.5 | Clustering Results

- We finally examine how the detected changes affect the spatial clustering of the fitted extremal dependence structure.
- For each fitted conditional extremes model, the aggregated dissimilarity matrix is used to partition the 40 stations into three clusters using the clustering procedure described in Section **??**.
- As the changepoint analysis provides no consistent evidence of a change in spring or autumn, we refit the conditional extremes model to the complete time series for these seasons.
- For summer, the pointwise tests identify a broader candidate change region during the late 1980s and early 1990s.
- We use 1990 as a representative division within this region and fit separate models to season-years 1960–1990 and 1991–2020.
- For winter, the three norms provide their clearest common evidence at season-year 1998.
- We therefore fit separate models to winter season-years 1960–1998 and 1999–2020.
- These divisions treat a candidate (t) as a change between season-years (t) and ($t + 1$). Confirm or update the selected divisions after completing the multiplicity-adjusted changepoint analysis.
- Figure 4 presents the resulting spatial clusterings.
- Although station coordinates and elevation do not directly determine the clustering assignments, the fitted clusters exhibit substantial spatial coherence.

- Neighbouring stations are frequently assigned to the same cluster, suggesting that the estimated extremal dependence structure contains a strong geographical component.
- The spring clustering broadly separates groups of stations along the northern and Mediterranean coasts (in blue) from groups in the western (orange) and central interior (red).
- Nevertheless, the boundaries are not determined solely by physical proximity, with some geographically separated stations assigned to the same dependence cluster.
- The autumn clustering exhibits a stronger broadly coastal–interior structure.
- Several stations along the northern and eastern coasts are grouped together, similar to the spring clustering, while the other two clusters comprise locations in the west/south-west and elsewhere, respectively.
- For summer, the two fitted periods retain some common large-scale structure.
- However, a number of stations change their clustering relationships after 1990, most visibly in northwestern Spain, along the southern coast and across parts of the western interior.
- The summer changepoint therefore corresponds to a reorganization of particular regional relationships rather than a complete replacement of the earlier clustering structure.
- The winter clustering, in contrast, changes more extensively across the 1998 division.
- In particular, the post-1998 fit produces a different grouping of many stations along the northern coast, in the western and central interior, and along the Mediterranean coast.
- This broader spatial reorganisation, as opposed to the more localised changes seen in summer, is consistent with the late-1990s signal being identified by all three norms in the changepoint analysis.

5 | DISCUSSION

ACKNOWLEDGMENTS

This is acknowledgment text Elbaum, Malishevsky, and Rothermel (February 2002). Provide text here. This is acknowledgment text. Provide text here. This is acknowledgment text. Provide text here. This is acknowledgment text. Provide text here. This is acknowledgment text. Provide text here. This is acknowledgment text. Provide text here. This is acknowledgment text. Provide text here. This is acknowledgment text. Provide text here. This is acknowledgment text. Provide text here.

Author contributions

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Financial disclosure

None reported.

Conflict of interest

The authors declare no potential conflict of interests.

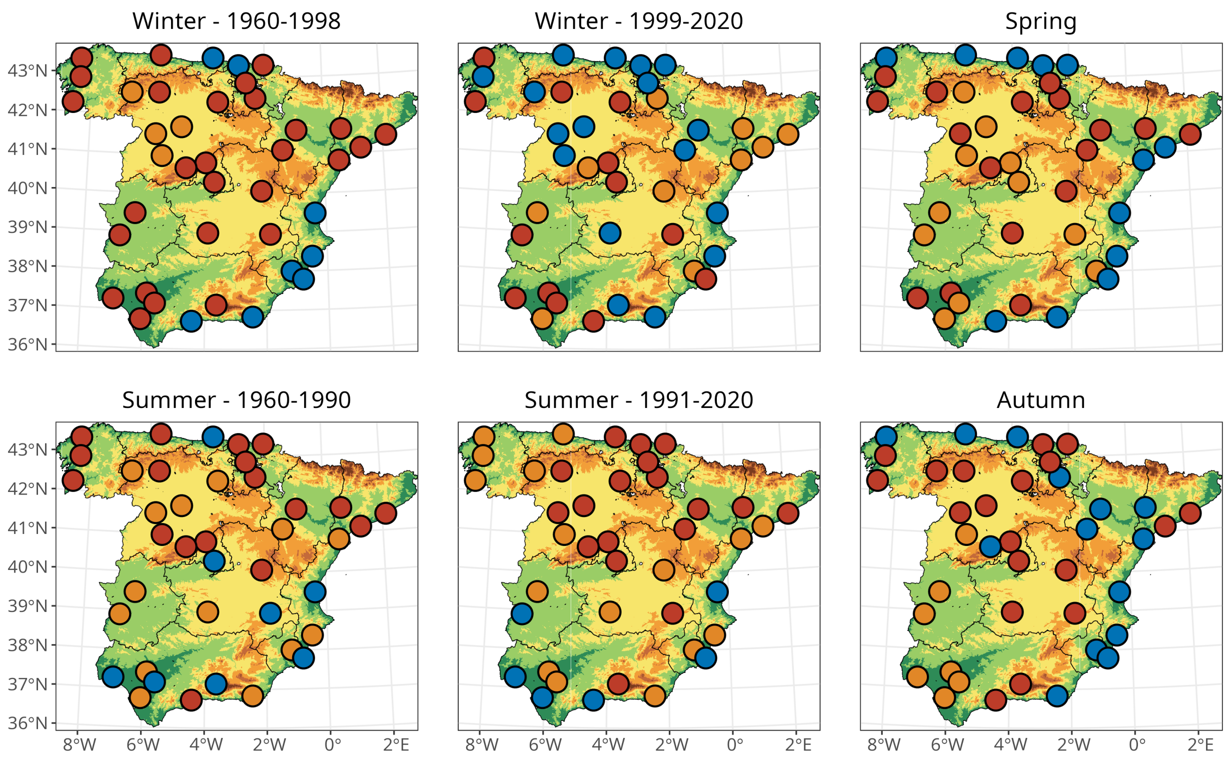

FIGURE 4 Spatial clustering of the 40 Spanish stations obtained from the fitted conditional extremes dissimilarity matrices. Spring and autumn are fitted using their complete time series. Summer is fitted separately for season-years 1960–1990 and 1991–2020, while winter is fitted separately for season-years 1960–1998 and 1999–2020. Three clusters are used in each fit. Colours distinguish cluster membership within each panel. Background shading represents elevation.

SUPPORTING INFORMATION

The following supporting information is available as part of the online article:

How to cite this article: Williams K., B. Hoskins, R. Lee, G. Masato, and T. Woollings (2016), A regime analysis of Atlantic winter jet variability applied to evaluate HadGEM3-GC2, *Q.J.R. Meteorol. Soc., 2017;00:1–6*.

APPENDIX

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